



Miss Mehwish: 03171491481

CS302p

Final Term (Live Quiz)

PAID VU LMS HANDLING by Mam Mehwish

03184148783

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Subject Code: CS302P

Question 1:

Three decade counter would have _____

- 3 BCD counters
- 4 BCD counters
- 2 BCD counters
- 5 BCD counters ☒

Question 2:

How many flip-flops are required to create a counter circuit which can count from 0 to 7?

- 1
- 3 ☒
- 2
- 4

Question 3:

Which of the following is the modified code of Excess-3 code?

- ASCII
- BCD
- Gray Code



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- EBDIC ☒

Question 4:

A counter circuit is usually constructed of _____

- A number of NOR gates connected in cascade form
- A number of flip-flops connected in cascade ☒
- A number of latches connected in cascade form
- A number of NAND gates connected in cascade form

Question 5:

An identity comparator is defined as a digital comparator which has _____

- No output terminals
- Three output terminals
- Two output terminals
- Only one output terminal ☒

Question 6:

Convert the binary number 1100 to Gray code.

- 0101
- 1000
- 0111
- 1010 ☒

Question 7:

Ripple counters are also called _____

- Synchronous counters
- SSI counters
- VLSI counters
- Asynchronous counters ☒

Question 8:

The parallel outputs of a counter circuit represent the _____



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- Parallel data word
- Counter modulus ☒
- Clock frequency
- Clock count

Question 9:

In a comparator, if we get input as $A > B$ then the output will be _____

- 0
- A
- B
- 1 ☒

Question 10:

Why do we use Gray Code?

- Invert the code
- Error Correction
- To count the number of bit changes ☒
- Error Detection

Question # 4 of 5

- An identify comparator is defined as a digital comparator which has
 - Only one output terminal
 - Two output terminals ✓
 - No output terminals
 - Three output terminals

Question # 5 of 5

- TTL 74LS85 is a
 - 1-bit digital comparator



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- ☐ 8-bit word comparator
- ☐ 4-bit magnitude comparator ✓
- ☐ 8-bit magnitude comparator

1. The parallel outputs of a counter circuit represent the:

- ☐ Counter modulus
- ☐ Clock frequency
- ☐ Parallel data word ✓
- ☐ Clock count

2. Ripple counters are also called:

- ☐ Asynchronous counters ✓
- ☐ VLSI counters
- ☐ Synchronous counters
- ☐ SSI counters

3. A decimal counter has:

- ☐ 10 ✓
- ☐ 8
- ☐ 16
- ☐ 4

4. How many select lines would be required for an 8x1 multiplexer?

- ☐ 8
- ☐ 6
- ☐ 3 ✓
- ☐ 2



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5. To create a Johnson Counter by using D-flip flops, how flip flops should be connected together?

- ☐ Q output of previous flip flop should be connected to clock input of second flip flop
- ☐ Q output of previous flip flop should be connected to D input of second flip flop ✓
- ☐ D input of second flip flop should be connected to clock input of the previous flip flop
- ☐ Q output of previous flip flop should be connected to D input of second flip flop

1. What is the maximum possible range of bit-count specifically in n-bit binary counter consisting of 'n' number of flip-flops?

- ☐ 0 to $2^n - 1$ ✓
- ☐ 0 to 2^n
- ☐ 0 to $2^n + 1$
- ☐ 0 to $2^n!$

2. How many flip-flops are required to create a counter circuit which can count from 0 to 7?

- ☐ 3 flip-flops ✓
- ☐ 4 flip-flops
- ☐ 5 flip-flops
- ☐ 6 flip-flops

3. In 8x1 multiplexer, which one of the following lines must always be connected to ground in order to activate the multiplexer?

- ☐ W
- ☐ Y
- ☐ G' ✓
- ☐ Vcc



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4. In 8x1 multiplexer, which one of the following lines must always be connected to ground in order to activate the multiplexer?

- ☐ W
- ☐ Y
- ☒ G' ✓
- ☐ V_{cc}

5. To create a Johnson Counter by using D-flip flops, how flip-flops should be connected together?

- ☒ Q' output of previous flip flop should be connected to D input of second flip flop ✓
- ☐ Q output of previous flip flop should be connected to clock input of second flip flop
- ☐ Q output of previous flip flop should be connected to clock input of second flip flop
- ☐ Q output of previous flip flop should be connected to D input of second flip flop

6. A decimal counter has

- ☒ 10 ✓
- ☐ 20
- ☐ 15
- ☐ 5

cs302p

Following diagram is of _____ code conversion.

ASCII

Gray

BCD



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Excess-3 ☒

Which of the following is an invalid BCD code?

1001

0101

0011

1101 ☒

TTL 74LS85 is a _____

1-bit digital comparator

8-bit word comparator

8-bit magnitude comparator ☒

4-bit magnitude comparator

An identity comparator is defined as a digital comparator which has _____

Only one output terminal

Two output terminals ☒

No output terminals

Three output terminals

A decimal counter has _____ states.

9

7

10 ☒

2



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How many flip-flops are required to create a counter circuit which can count from 0 to 7?

3 ☒

1

2

4

Three decade counter would have _____

4 BCD counters

3 BCD counters ☒

5 BCD counters

2 BCD counters

A counter circuit is usually constructed of _____

A number of NOR gates connected in cascade form

A number of NAND gates connected in cascade form

A number of flip-flops connected in cascade ☒

A number of latches connected in cascade form

Why do we use Gray Code?

To count the number or bit changes

Error Correction

Invert the code

Error Detection ☒

Which one is a basic comparator?



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NAND

AND

XOR ☒

ORCS302p

Question # 1

The K-map based Boolean reduction is based on the following Unifying Theorem: $A + A' = 1$.

Force

Non Impact

Impact

Complementarity ☒

Question # 2

Don't care conditions can be used for simplifying Boolean expressions in _____.

Registers

Latches

K-maps ☒

Terms

Question # 3

A Karnaugh map (K-map) is an abstract form of _____ diagram organized as a matrix of squares.

Block diagram

Venn Diagram ☒



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Triangular Diagram

Cycle Diagram

Question # 4

When implementing a half adder by using gates, which of the following gates is used to give Sum (S) as an output?

OR

NOR

XOR ☒

AND

Question # 5

Total number of inputs in a half adder is _____.

3

1

4

2 ☒

Question # 6

When implementing a half adder by using gates, which of the following gates is used to return carry bit as an output?

NAND

XOR

OR

AND ☒

Question # 7



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There are _____ cells in a 5-variable K-map.

24

64 ☒

32

16

Question # 8

Which of the following gates is used for parity generation for a binary data string?

OR

NOR

XOR ☒

AND

Question # 9

To build a full adder using gates, how many gates are required?

2 AND gates, 3 OR gates, 1 XOR gates

1 AND gate, 2 OR gates, 1 XOR gate

2 AND gates, 2 XOR gates, 1 OR gate ☒

2 AND gates, 2 OR gates, 2 XOR gates

Question # 10

Each product term of a group, $w'xy'$ and wy , represents the _____ in that group.

Sum of Maxterms

POS ☒

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Input

Sum of Minterms

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